

Question	Answer	Mark	Mark scheme	Additional guidance
16 (a)	Proof	M1 M1 C1	for expansion of $(2m + 1)^2$ or $(2n - 1)^2$, all 4 terms correct with or without signs (and no additional terms) or 3 out of 4 terms correct with signs, eg $4m^2 + 2m + 2m + 1$ or $4n^2 - 2n - 2n + 1$ or for correct expansion of $4(m + n)(m - n + 1)$ or $(m + n)(m - n + 1)$ eg $4m^2 - 4mn + 4m + 4mn - 4n^2 + 4n$ oe or $m^2 - mn + m + mn - n^2 + n$ oe or for $[2m + 1 + 2n - 1][(2m + 1) - (2n - 1)]$ for correct expression after expansion for $(2m + 1)^2 - (2n - 1)^2$ eg $(4m^2 + 4m + 1) - (4n^2 - 4n + 1)$ or $4m^2 + 4m + 1 - 4n^2 + 4n - 1$ oe (= $4m^2 + 4m - 4n^2 + 4n$) or for $[2m + 1 + 2n - 1][2m + 1 - 2n + 1]$ for a complete proof without any errors, eg uses difference of two squares to show that LHS = RHS or expands both sides and shows that LHS = RHS or expands and simplifies LHS and factorises convincingly to get RHS	Note that, for example, $4m + 1$ is regarded as 3 terms in the expansion of $(2m + 1)^2$ Must see correct expression $\begin{aligned} &4m^2 - 4n^2 + 4m + 4n \\ &= 4[(m^2 - n^2) + (m + n)] \\ &= 4[(m + n)(m - n) + (m + n)] \\ &= 4(m + n)(m - n + 1) \end{aligned}$
(b)	Yes (supported)	C1	for yes with explanation, eg $2m + 1$ and $2n - 1$ are odd numbers (for any positive integer value of m, n) and the right-hand side is a multiple of 4	